International Journal of Cognitive Computing in Engineering 4 (2023) 6–20

Contents lists available at ScienceDirect

International Journal of Cognitive Computing in Engineering

journal homepage: https://www.keaipublishing.com/en/journals/international-journal-of-cognitive-computing-in-engineering/

# Comparison of multiclass classification techniques using dry bean dataset

Md Salauddin Khan[a,*], Tushar Deb Nath[b], Md Murad Hossain[c], Arnab Mukherjee[d], Hafiz Bin Hasnath[a], Tahera Manhaz Meem[a], Umama Khan[e]

[a] *Statistics Discipline, Science, Engineering and Technology School, Khulna University, Khulna-9208, Bangladesh*
[b] *Department of Computer Science and Engineering, Jahangirnagar University, Dhaka, Bangladesh*
[c] *Modeling and Data Science, University of Turin, Italy*
[d] *Mathematics Discipline, Science, Engineering and Technology School, Khulna University, Khulna-9208, Bangladesh*
[e] *Biotechnology and Geneetic Engineering Discipline, Life Science School, Khulna University, Khulna-9208, Bangladesh*

ARTICLE INFO

*Keywords:*
Dry bean
Interquartile range
ADASYN
Multiclass classification techniques
Performance measures

ABSTRACT

*Background:* The application of classsification methods through multivariate and machine learning techniques has enormous significance in agricultural sector. It is vital to classify various types of seeds as well as identify the quality of seeds which has a great impact on the production of crops. There is a wide range of genetic variations in dry beans all over the world. Many studies have been conducted previously on various dataset to indentify the sorts of dry beans, however most of them focused on machine learning techniques with binary classification.
*Objective:* The aim of this study is to investigate a reliable classifier which has the lowest noise implications and establish an algorithm for dry bean classification effectively. This paper focuses on outlier removals, oversampling with Adaptive Synthetic (ADASYN) algorithm and finding the best classifier to guarantee the highest possible accuracy.
*Methods:* The raw dataset for this study was accessed from UCI Machine Learning Repository. The dataset contained grains having 16 features, 12 dimensions, and 4 distinct shapes. For the purpose of eliminating missing values from the dataset, interquartile range (IQR) with python programming was utilized. Eight most popular classifiers were used in this study which are Logistic Regression (LR), Naïve Bayes (NB), k-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB), Support Vector Machine (SVM), and Multilayer Perception (MLP) with balanced and imbalanced classes. The authors utilized frequency tables, bar diagrams, boxplots, analysis of variance for descriptive analysis as well as data preprocessing.
*Results:* The XGB classifier preferably outperformed than other classifiers with balanced and imbalanced distribution of dry beans within each class. It has acquired accuracy (ACC) 93.0% and 95.4% in imbalanced and balanced classes respectively. In case of balanced dataset, after application of ADASYN algorithm both KNN and RF techniques also performed well regarding the Classification Accuracy (ACC), Sensitivity (SE), Specificity (SP) and Cohen's kappa coefficient (Kappa) etc. The most important attributes for classifying the dry beans were found ShapeFactor2, Minor Axis Length, and ShapeFactor1 along with EquivDiameter, Roundness and ConvexArea.
*Conclusions:* For classification of dry seeds, the XGB classifier had performed well when the dataset contained both balanced and imbalanced distribution in classes. Also, it is the primary approach of identifying the classes of seeds/beans with balanced or not. If the classes of the target variable are balanced well, then the KNN and RF algorithms may be applied along with XGB technique for more accurate classification.

## 1. Introduction

Classification techniques are becoming more popular in the fields of medical, biostatistics, bioinformatics, agriculture, business etc. as machine learning applications. Machine learning is a subfield of artificial intelligence that enables computers to understand from existing data and estimate the existence of unidentified targets (Yahyaoui, A. & Yumuşak, N., 2018). Many scientists have worked to develop and identify qualityful seeds in agricultural sector by applying artificial intelligence algorithms with a view to ensuring food security. Also, several studies have been conducted to detect the quality of dry beans using various machine learning techniques.

There is a variety of computational equipments available for controlling the quality of foods and agricultural goods. But most of them are done with the use of conventional techniques. For example, seed categorization is carried out usually based on human understanding and de-

* Corresponding author.
*E-mail address:* salauddinstat@ku.ac.bd (M. Salauddin Khan).

https://doi.org/10.1016/j.ijcce.2023.01.002
Received 10 March 2022; Received in revised form 4 January 2023; Accepted 7 January 2023
Available online 14 January 2023

termining the type of dry beans requires skillful person and takes huge time manually. When the array of seeds appears so similar, manually categorizing them becomes a challenging process. Even, it is almost impossible for a human operator to interpret or handle such seeds except specific tools or automatic software procedures (Mendoza and Aguilera, 2010; Liu *et al.*, 2011; Savakar, 2012; Rodríguez-Pulido *et al.*, 2013; Gómez-Sanchis *et al.*, 2012, Stegmayer *et al.*, 2013; Khatri *et al.*, 2022). In today's world, the inspection of the quality of seeds, fruits and vegetables along with examination and categorization of seeds and grains have been performed worldwide to meet these demands with help of machine learning and computer vision. The purpose of seed categorization is to ensure high-quality food product in greater quantities.

The dry bean (*Phaseolus vulgaris L.*) is the most nutritious and widely cultivated vegetable found (Fabaceae-Leguminosae) all over the world. The purification of dry beans play an important role in the economy of agriculture based countries like Bangladesh, India, Pakistan etc. throughout the winter season. Unfortunately, deterioration in seed quality may begin at any point in the plant's development stage from fertilization onward due to the effects of changing climate and other environmental factors. Breeding new seed cultivars and determining their traits, which are the significant variables for growth of plants properly and may improve the response of plants and/or tolerance to environmental stimuli (Ceyhan *et al.*, 2012). The process of seed identification is time-consuming and may be interpreted in a variety of ways. From the practical point of view, it becomes more challenging with respect to commercial and technical aspects. Specially, various dry bean species tend to vary in color and the geometrical data carry no information about the bean color. For this reason, it is crucial not only economically but also technically to build an automated technique to detect as well as categorize seed features rapidly and repeatedly (Granitto *et al.*, 2002; Bacchetta *et al.*, 2008).

In the perspective of cultivation, the qualities of seed influence the crop production greatly. In the recent years, knowledge-based technologies such as statistical learning, fuzzy logic and artificial neural networks (ANN) have been used in inspection, classification, prediction and segmentation of food product quality (León-Roque *et al.*, 2016; Du & Sun, 2004). The combination of Computer Vision Systems (CVS) and ANN produce a potent machine vision inspection tool. Many researchers have employed machine learning algorithms to evaluate the quality of beans using various analytical techniques (León-Roque *et al.*, 2016; Lawi, A. & Adhitya, Y., 2018). Random forest (RF) is an ensemble learning technique that compares preferably well with or outperforms various classification algorithms including SVM, C4.5, AdaBoost, KNN, LR, Stochastic Gradient Boosting Trees, Extreme Learning Machine, Sparse Representation-Based Classification, and Deep Learning (Breiman, 2001; Zhang *et al.*, 2017; Barbon et al., 2016). Most of the studies follow traditional approaches to predict seed types. For instance, SVM (Subasi, 2015; Yahyaoui *et al.*, 2018), RF (Barbon et al., 2016), KNN, NB, DT and MLP (Koklu *et al.*, 2020) algorithms have showed better performance in solving classification problems in a variety of agricultural fields.

The aim of this study is to utilize a set of classification techniques and find out the best classifier which identifies the actual types of dry beans. In addition, ADASYN algorithm was adopted to make distribution of classes equally; and also employed boxplot and Interquartile Range (IQR) for removing outliers that improve slightly classification performance. Further, our investigation demonstrates that the XGB classifier may be applied in lieu of widely used KNN, RF and LR techniques.

The present study offers the following contributions:

i) ***Data processing:*** The core contribution of this study has been described in data prepocessing and classification stages including data scaling, outliers removing, and applying ADASYN algorithms to eliminate the issues occurred due to imbalanced class distribution of dataset.

ii) ***Classification of balanced and imbalanced dataset***: Nowadays the researchers employ the traditional classifiers namely Linear Regression (LR), Naïve Bayes (NB), k-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB), Support Vector Machine (SVM), and Multilayer Perception (MLP) with balanced and imbalanced data to provide comparison with state-of-the-art algorithms.

iii) ***Performance evaluations:*** In experimental evaluation, the performances of the mentioned classifiers are compared and evaluated in terms of Accuracy (ACC), Sensitivity (SE), Specificity (SP), False Positive Rate (FPR), Cohen's Kappa Coefficient (Kappa), $F_1$- score, Mean Square Error (MSE) and Area Under the Curve (AUC).

iv) ***Comparison of classifiers:*** There are no comparative studies on multiclass classification techniques and data distribution which address subject-wise problems. From experimental evaluation, the XGB classification technique shows better performance among the selected classifiers with the help of ADASYN algorithms. Herein it has improved accuracies of 1.4%, 5.5%, 0.40%, and 11.5% compared to RF, DT, KNN, MLP respectively.

v) ***Statistical evaluation:*** The Receiver Operating Characteristic (ROC) curves are constructed by plotting the True Positive Rate (TPR) against False Positive Rate (FPR) at various threshold settings to diagnose the ability of the classifiers and validation of the proposed ML-based system.

The rest of the paper is outlined as follows. Section 2 reviews the applications of machine learning algorithms in seed identification. In Section 3, materials and research framework are described including data sources, descriptive statistics of the variables, data preprocessing stage and different classification techniques. The performance measures are also mentioned in this section. Section 4 presents the experimental results, and also shows graphical and statistical performances with the help of confusion matrix. A brief discussion is provided in Section 5 and finally, draws the conclusion of the work in Section 6.

## 2. Related works and motivations

This section provides a brief discussion about recent works which are related to the classification of different seed varieties. Almost all the existing machine leaning (ML) algorithms have been used with various morphological, tonal, textural and color features for seed classification.

Oliveira *et al.* in 2021 developed a fast and reliable computer vision system to classify fermented cocoa beans into four categories (Oliveira *et al.*, 2021). To identify the samples, hand-crafted characteristics were extracted from the beans as predictors. They used RF to determine the quality of fermented beans and proposed it as a cut-test by using digital Red, Green and Blue (RGB) imagess. Sanl *et al.* evaluated the performance on different datasets using KNN, J48, SMO, NB, NBM, BAGGING and JRIP classification algorithms (Sanlı *et al.*, 2020). A machine learning technique was proposed by Islam *et al.* to identify illness in potato plants using leaf images. Their study obtained 95% accuracy in classifying illness in potato using an SVM on over 300 samples. Their approaches enabled the widespread diagnosis of plant diseases by automated detection. However, the seriousness of the identified ailments has not yet been established (Islam *et al.*, 2017). Gürcan *et al.* analyzed Turkish literature using supervised machine learning methods with variety of factors (Gürcan, F., 2018). On Turkish news texts, the authors compared the performance of Multinomial NB, Bernoulli NB, SVM, KNN and DT algorithms. Another study classified hazardous online activities into different categories using J48, PART and SVM (Goseva-Popstojanova *et al.*, 2012). The authors made an attempt to differentiate between various types of malicious activities directed towards internet platforms.

Koklu *et al.* introduced another Computer Vision System (CVS) for recognizing registered varieties of dry beans with comparable traits in order to get consistent seed types from crop output (Koklu *et al.*, 2020). They assessed their performance by comparing MLP, SVM, KNN, and DT classification algorithms using 10-fold cross validation strategy. However, the authors considered two-dimensional images of bean varietals to estimate their shape and size. An ML based approach was adopted to extract features from the samples (Sáez *et al.*, 2015). Performance was evaluated for a variety of nominal classification techniques including LR and ANN. Additionally, the authors developed a set of ordinal classification strategies for the three-class issues. Andhalkar *et al.* have solved multiclass problems by reducing them to multiple binary classification problems utilizing the OVA (one versus all) and OVO (one versus one) approaches (Andhalkar S. & Momin B. F., 2018). The authors developed a novel algorithm using hybrid technique with addressing data imbalances.

Lang *et al.* introduced a robust and fast technique to minimize the number of training parameters. But the researchers did not provide any direction for a multilayer scenario when deals with multiclass problems concurrently (Lang *et al.*, 2016). Slowinski investigated a data collection using machine learning and deep learning algorithms for autonomously categorizing bean species (Słowiński, G. 2020). The author used Multinomial Bayes, SVM, DT, RF, voting classifier and ANN for visualization and performance analysis. A unified technique for multiclass object recognition was established to address structural label prediction in pictures by (Desai et al., 2011). The authors characterized parameter estimation as a problem of maximum margin learning. They proposed another object recognition system with spatial interactions that could be swiftly learnt in an end-to-end racist and discriminating way.

A hybrid model was developed using machine learning tools for disease prediction (Rathi *et al.*, 2016). The researchers looked at how a machine learning system predicted the target class based on stated characteristics. They examined the accuracy, root mean square error, mean absolute error, Kappa statistics, sensitivity and specificity of four distinct classifiers utilizing the MRMR feature selection approach and four different classifiers including SVM, Function Tree, End Meta and NB. Rehman *et al.* presented an overview of studies in the fields of agriculture as well as forest that use statistical machine learning techniques. The authors addressed the shortcomings of earlier algorithms for agricultural applications and focused on the most successful approaches for these sectors (Rehman *et al.*, 2019).

In the fields of agriculture, the conventional CVSs have been utilized widely in seed identification and classification under the controlled conditions. Most of the existing systems performed well under high-quality captured image of seeds but classification performance substantially decreases with various levels of illumination and increasing number of seeds. To address these issues, Alzubi *et al.* introduced various machine learning approaches with SVM classifier and Harris Hawks optimization algorithm (Alzubi *et al.*, 2021), ANN with various Metaheuristic algorithms (Movassagh *et al.*, 2021), dynamic programming-based ensemble design algorithm (Alzubi *et al.*, 2020), ANN and NB classifiers (Sethuraman *et al.*, 2019) and search engine optimization technique (Alzubi *et al.*, 2019).

Recently, Kumari & Rai (Kumari et al., 2021; Rai et al., 2022a) introduced an enhanced approach using ensemble of several machine learning algorithms and focused on the average probability of every individual classifier. Whereas the prediction probabilities are weighted by the importance of classifier and summed up. Then the target label with the greatest sum of weighted probabilities wins the vote. The performance of proposed ensemble soft voting classifier has attained higher results as compared to baseline classifiers (Muralidharan et al., 2021; Kalaivani et al., 2022; Islam et al., 2022).

From the above discussions, it is clear that these traditional approaches still have various limitations in real-world scenarios. Also, there have been very few comparative studies on classification technique where the subject-wise problems were addressed in different sorts of seeds. To consider the challenges, this work introduces a novel machine learning algorithm that is capable of solving the multiclass problems of subject heterogeneity by incorporating dimensional and shape features with machine learning classifier.

**Table 1**
Sample distribution of the types of dry beans.

| No. | Name | Piece | Seed Weight (Average gram per seed) |
|---|---|---|---|
| 1 | Seker | 2027 | 0.49 |
| 2 | Barbunya | 1322 | 0.76 |
| 3 | Bombay | 522 | 1.92 |
| 4 | Cali | 1630 | 0.61 |
| 5 | Horoz | 1928 | 0.52 |
| 6 | Sira | 2636 | 0.38 |
| 7 | Dermason | 3546 | 0.28 |
| **Total** | | **13,611** | **0.71** |

## 3. Research framework and relevant materials

The proposed framework consists of multiple stages as shown in Fig. 1. Data collection is the primary stage of this work which is introduced as data acquisition. Statistical method IQR and balancing method ADASYN are used for checking outliers and balancing frequncy of each class respectively. The data are scaled in the next stage and separated into test and training datasets. In this research, eight known classifiers LR, RF, DT, SVM, XGB, KNN, NB and MLP are utilized to classify the dry beans. All the algorithms are implemented using python program to determine optimized parameters by tuning. The measures like AUC, ACC, MSE, $F_1$-score, FPR, Kappa, SE and SP from confusion matrix are computed to evaluate the performance of the classifiers.

### *3.1. Data collection*

For experimental evaluation, the raw dataset was collected from the University of California, Irvine's Machine Learning Repository in this study. The dataset is available in the following link: https://archive.ics.uci.edu/ml.Dry+Bean+Dataset that was extracted by Koklu *et al.* from different images of dry beans. The dry bean dataset comprised of 13,611 items of seven distinct registered dry beans acquired by a high-resolution camera for multiclass classification (Koklu *et al.*, 2020).

The collected dataset has sixteen distinct features, twelve dimensions and four distinct shapes. 1) Area (A): the area of the bean zone and the number of pixels included inside its limits; 2) Perimeter (P): the circumference of a bean is defined as its border length; 3) Length of the major axis (L): the distance between the endpoints of the longest line that can be drawn from a bean. 4) Minor axis length (l): the longest line that can be drawn perpendicular to the main axis from the bean; 5) Aspect ratio (K): this parameter establishes the link between L and l; 6) Eccentricity (Ec): Eccentricity of the ellipse that shares the region's moments; 7) Convex Area (C): the number of pixels contained inside the smallest convex polygon capable of containing the area of a bean seed. 8) Equivalent diameter (Ed): the diameter of a circle whose area is equal to that of a bean seed. 9) Extent (Ex): the ratio of pixels included inside the enclosing box to the area of the bean; 10) Solidity (S): Another term for convexity. The ratio of convex shell pixels to those present in beans; 11) Roundness (R): Calculated using the formula: $(4\pi A/P^2)$; 12) Compactness (CO): Measures the roundness of an object: Ed/L; 13) ShapeFactor1 (SF1); 14) ShapeFactor2 (SF2); 15) ShapeFactor3 (SF3); and 16) ShapeFactor4 (SF4). Dry bean class was considered as response variable. Here, Seker, Barbunya, Bombay, Cali, Dermason, Horoz, and Sira were the dry bean classes.

### *3.2. Basic information and descriptive analysis*

The frequency distribution of dry bean classes together with the total number of pieces within each class and the average weight of each class is shown in Table 1 and Fig. 2. Weight diversity existed due to seed weight variations as samples are acquired from uniformly weighted

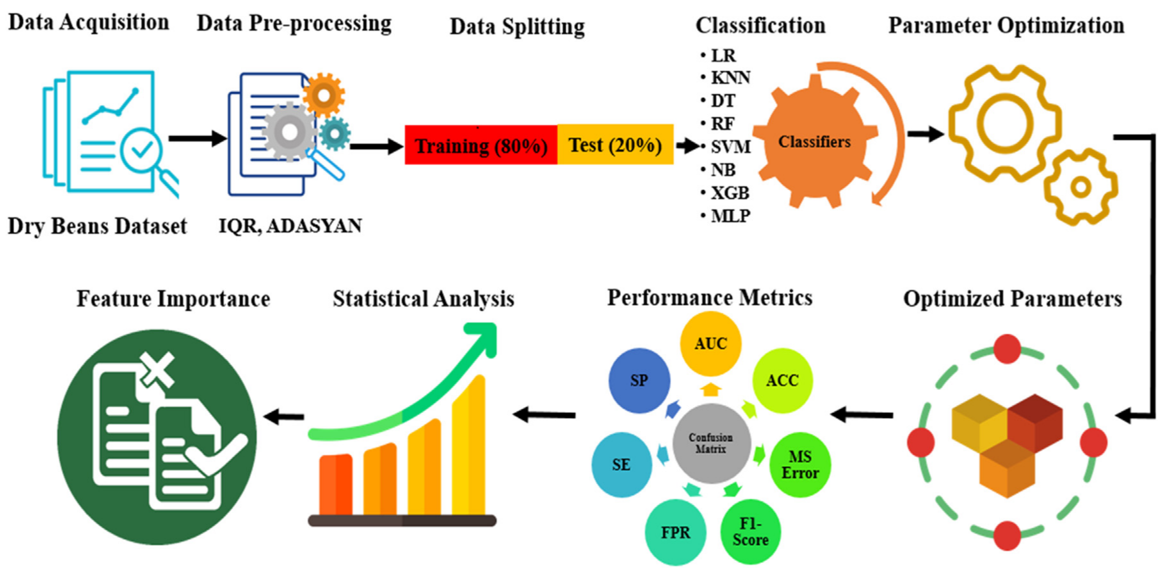


**Fig. 1.** The overall workflow of the study.

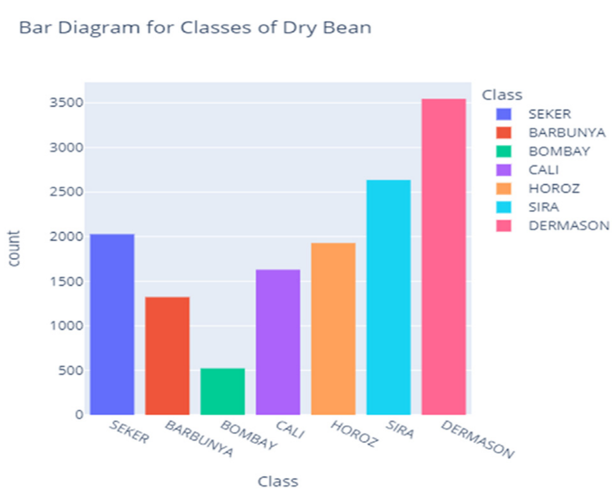


**Fig. 2.** Frequency distribution of different dry beans classes.

**Table 2**
Statistical distribution for the selected features of dry bean (in pixels).

| No | Features | Min. | Max. | Mean | Std. |
|---|---|---|---|---|---|
| 1 | Area | 20,420 | 254,616 | 53,048.28 | 29,324.09 |
| 2 | Perimeter | 524.74 | 1985.37 | 855.28 | 214.28 |
| 3 | Major Axis Length | 183.60 | 738.86 | 320.14 | 85.69 |
| 4 | Minor Axis Length | 122.51 | 460.19 | 202.27 | 44.97 |
| 5 | Aspect Ratio | 1.025 | 2.430 | 1.583 | 0.247 |
| 6 | Eccentricity | 0.219 | 0.912 | 0.751 | 0.092 |
| 7 | Convex Area | 20,684 | 263,261 | 53,768.20 | 29,774.91 |
| 8 | Equiv Diameter | 161.24 | 569.37 | 253.06 | 59.17 |
| 9 | Extent | 0.555 | 0.866 | 0.749 | 0.049 |
| 10 | Solidity | 0.919 | 0.995 | 0.987 | 0.005 |
| 11 | Roundness | 0.489 | 0.991 | 0.873 | 0.059 |
| 12 | Compactness | 0.641 | 0.987 | 0.799 | 0.062 |
| 13 | Shape Factor 1 | 0.003 | 0.011 | 0.007 | 0.001 |
| 14 | Shape Factor 2 | 0.001 | 0.004 | 0.002 | 0.001 |
| 15 | Shape Factor 3 | 0.410 | 0.975 | 0.644 | 0.099 |
| 16 | Shape Factor 4 | 0.948 | 0.999 | 0.995 | 0.004 |

seeds of each type. The Bombay class has the lowest number of seed samples (522) and the highest average seed weight (1.92g per seed). Dermason has the most observations (3546) with an average weight 0.28. The study (Koklu, *et al.*, 2020; Paliwal *et al.*, 2001) identified some relevant dimension features for seed separation. As a result, the samples of seven classes were imbalanced. To balance the dry bean classes, the ADASYN approach is applied and different machine learning methods are used to evaluate classification performance. Table 2 illustrates the statistical distribution of characteristics of dry bean variety, including minimum (Min.), maximum (Max.), mean (Mean) and standard deviation (Std.).

### *3.3. Data pre-processing*

In this subsection, we have detected the missing values and outliers from the dataset using statistical method boxplot and interquartile range (IQR). Though there are no missing values, the outliers are in several categories of dry bean dataset. Most of the variables in Fig. 3 has a higher proportion of outliers including Area, Perimeter, Minor Axis Length, Eccentricity, Convex Area, EquivDiameter and ShapeFactor4. Fig. 3 illustrates the boxplot for each variable. The center of horizontal line represents the median and the vertical line shows the first quartile ($Q_1$) and third quartile ($Q_3$) from bottom to top respectively. The lower and upper fences lie in the first and third quartiles and are computed as ($Q_1$-1.5 × IQR) and ($Q_3$+1.5 × IQR) respectively. Outliers have been shown as vertically dot points on the outside of the lower and upper fences. These outliers are removed with the help of IQR in python program to the related variables. The variables are found after removing outliers as shown in Fig. 4.

### *3.4. Classification models*

In this study, the well-known classifiers namely LR, NB, KNN and DT are followed. In contrast, state-of-the-art classifiers like as RF, XGB, SVM and MLP considerably outperform accuracy in a wide range of classification tasks including dry bean classification. To improve accuracy rate, each classifier contains a collection of hyperparameters that might have been adjusted.

#### *3.4.1. Logistic regression classifier (LR)*

The Logistic Regression (LR) model is a probabilistic statistical classification technique (Paliwal et al., 2001; Awad, M. & Khanna, R., 2015) used to build a categorical dependent variable or a categorical outcome variable. It uses one or more independent variables to predict binary or multiple responses to a categorical dependent variable (Igual, L. & Seguí, S., 2017). As the confined model makes assumption about under-

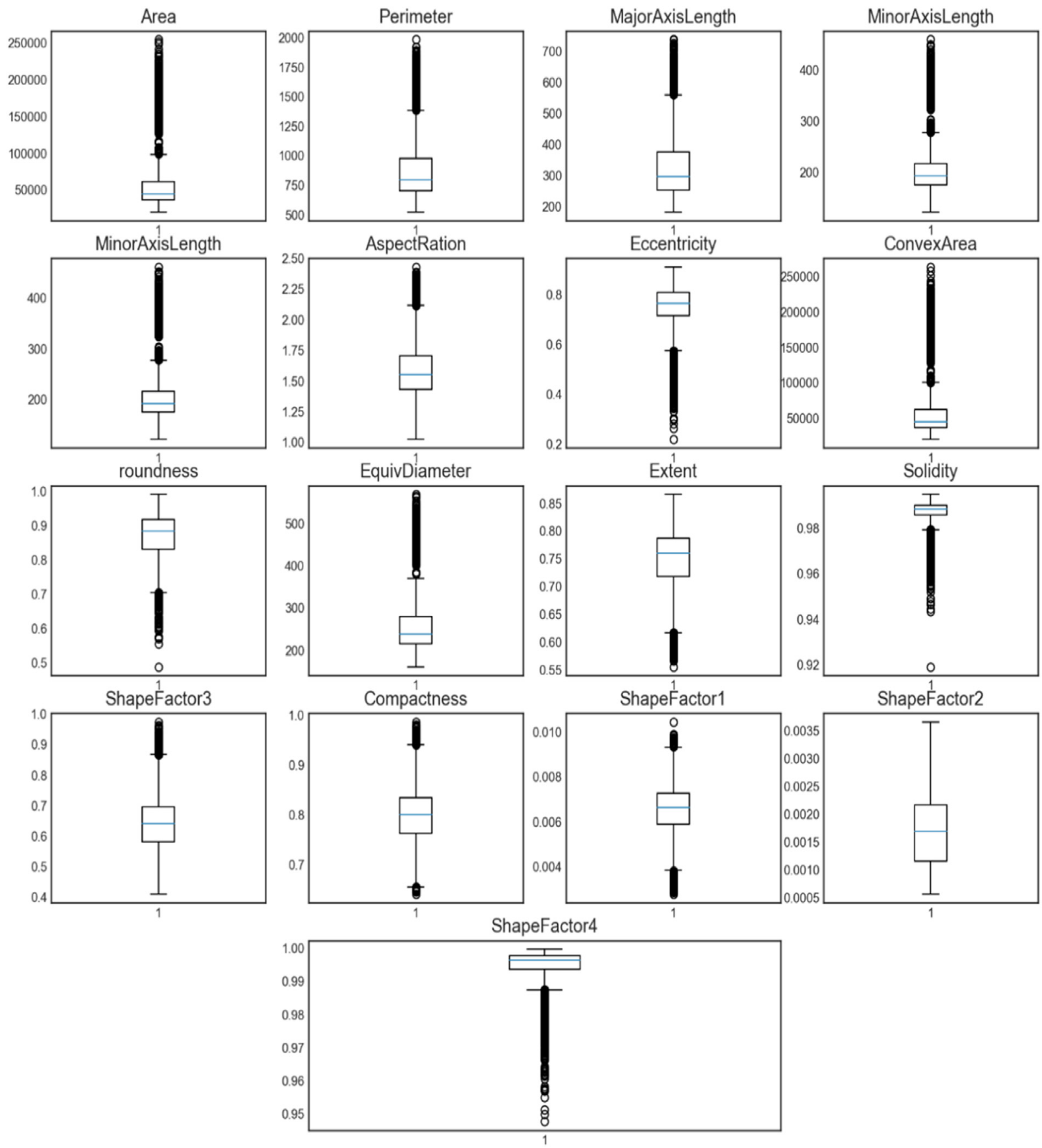


**Fig. 3.** Box-plot for different features of dry beans.

lying data distribution including predictor independence and a meaningful variable related with outcome variable. These assumptions must be satisfied before estimating the model (Agresti, A., 2002).

### 3.4.2. K-nearest neighbor classifier (KNN)

KNN is a distance based supervised machine learning technique that makes the use of training data to categorize new data points. It is used to solve classification and regression problems (Mukherjee *et al.*, 2022). It returns an integer number representing the productivity (labels) of a classification algorithm output. KNN is a memory-based classifier that reminds all training data points in order to predict test data by comparing an input sample to each training instance. It considers k training neighbors $x_r$ where $r = 1, \ldots, k$ that are closest to $x_0$ in terms of distance. For a given new data point $x_0$, the algorithm labels them based on a majority vote among the $k$ neighbors (Hastie *et al.*, 2009;Mukherjee *et al.*, 2021). It employs the KNN technique to run respective times with various values of $k$ and selects $k$ that minimizes the number of errors appropriately.

### 3.4.3. Decision tree classifier (DT)

One of the simple and straightforward machine learning algorithms is the Decision Tree (DT) classifier, which is based on the divide and conquer principle (Igual, L. & Seguí, S., 2017). A DT with internal nodes representing tests (on input patterns) and leaf nodes representing categories (of patterns) provides a class number (or output) to the pattern by filtering it through the tree tests. Each test provides conclusive and mutually exclusive results.

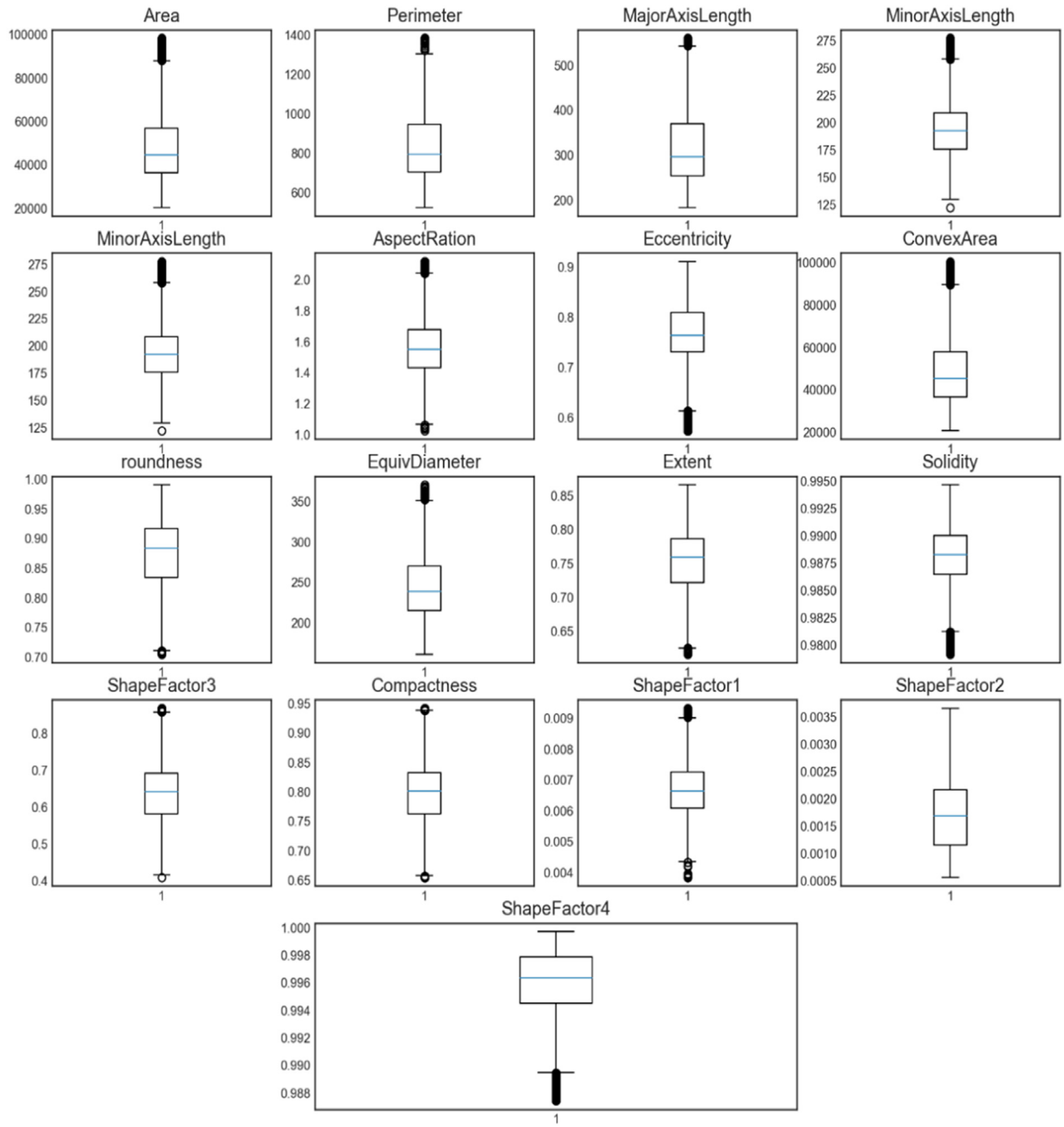


**Fig. 4.** Box-plot for the features after removing outliers engineering of the dataset.

### 3.4.4. *Random forest classifier (RF)*

In RF method, the number of trees and their maximum depth *i.e.*, evaluation of interactions are hyper-parameters. The RF (Awad, M. & Khanna, R., 2015) is a classification technique that provides a large number of de-correlated DTs. To develop the RF method in Python, we have utilized a few numbers of DT and Gini as impurity index.

### 3.4.5. *Support vector machine classifier (SVM)*

SVM is the most widely used classification technique for predicting the class label of unknown sample based optimal decision boundary. The task of SVM algorithm is to find out an optimal decision boundary that separates the n-dimensional feature vectors into two classes with a hyper-plane. A kernel function is adopted to train the SVM model and transfer the feature vectors into a higher-dimensional space (Müller *et al.*, 2018; Madhu *et al.*, 2021). Then the machine learning issue is solved as a convex optimization problem (Awad, M. & Khanna, R., 2015; Burges, C. J., 1998). It is anticipated that any new observation will fit neatly into either of the categories depending on maximum marginal hyperplane. Support vectors are the data points closest to the hyperplane that separate the classes (Awad, M. & Khanna, R., 2015; Awal *et al.*, 2021b).

### 3.4.6. *Naïve Bayes classifier (NB)*

NB is a probabilistic classifier that applies Bayes theorem to achieve the highest level of performance (Awal *et al.*, 2021a). It considers that every feature is independent, has similar contribution to the target class and never interferes with each other. The classifier performs well on large datasets having high dimensionality. NB classifier is conducive to real-time applications in the field of classification and disease prediction and is not sensitive to noise.

**Table 3**
Confusion matrix and representation for multiple classes.

| | | Predicted | | | | |
|---|---|---|---|---|---|---|
| | | $C_1$ | $C_2$ | $C_3$ | … | $C_n$ |
| Actual | $C_1$ | $T_1$ | $F_{12}$ | $F_{13}$ | … | $F_{1n}$ |
| | $C_2$ | $F_{21}$ | $T_2$ | $F_{23}$ | … | $F_{2n}$ |
| | $C_3$ | $F_{31}$ | $F_{32}$ | $T_3$ | … | $F_{3n}$ |
| | … | … | … | … | … | … |
| | $C_n$ | $F_{n1}$ | $F_{n2}$ | $F_{n3}$ | … | $T_n$ |

### 3.4.7. Extreme gradient boosting multi-classifier (XGB)

XGB is the natural extension of decision tree that integrates several decision trees in determining the final output rather than depending on individual decision tree. It is applicable for supervised learning tasks including regression, classification and ranking. It also generates an estimated model utilizing a collection of weak estimating approaches. A phase method like the other boosting procedures that generalizes the model by allowing optimization of a random differentiable loss function. 'Boosting' is a tree-generation approach that follows gradient descent to create new strong trees from existing ones. It directs the target function in the shortest possible path (Zhang *et al.*, 2017).

### 3.4.8. Multilayer perception classifier (MLP)

MLP is a fully connected feed forward ANN that learns from the pre-trained model and decides how to react for a new query. The deep learning classifier works well with over-fitting and under-fitting due to training with back propagation technique. It is able to handle stochastically the problems occurred in imbalanced data, which often allows approximate solutions for more complicated problems like fitness approximation. MLP consists of three connected layers of nodes to make equivalent conclusions about similar problems. MLP is made up of artificial neurons that are coupled in a hierarchical manner and work in collaboration (Koklu *et al.*, 2020; Przybył *et al.*, 2018).

### 3.5. Performance measures

There are several evaluation metrics to measure the performance of a machine learning algorithm. The effectiveness of a ML algorithm is determined by the percentage of correct predictions made among all predictions. The reported metrics are derived from confusion matrix. Confusion matrix is one of the best reliable measure techniques to describe the performance of a classifier against a set of known test data. True positive, true negative, false positive and false negative are used to construct confusion matrix. Table 3 shows the multiclass confusion matrix with predicted and actual class that is utilized for visualizing the performance of each class. The performance metrics like Accuracy (ACC), Error Rate, Sensitivity (SE), Specificity (SP), Mean Square Error (MSE), Recall, False Positive Rate (FPR), Kappa and $F_1$-score are employed to evaluate accurate predictions for classification problems as enlisted in Table 4 (Li *et al.*, 2021; Islam et al., 2022).

### 3.5.1. Receiver operating characteristics (ROC) curve

In machine learning, the graphical analyses are essential for performance evaluation when multiclass classification problems arrive. The ROC curve is plotted to depict the performance of the multiclass classifiers. Their Area Under Curve (AUC) is also computed to measure discriminative ability or how well it works in a particular clinical setting (Ahmed et al., 2021; Khan et al., 2022; Kumari et al., 2021; Rai et al., 2022b; Islam et al., 2022). In the ROC curve, AUC is equal to the probability and provides an aggregated measure. The values of AUC around 1 show that a model is excellent indicating that it has a high degree of separability. Alternatively, the values of AUC near 0 indicate lower performance model. The test is more visually effective if the ROC curve is nearer to the top left corner.

### 3.5.2. Graphically distribution of characteristics of the features

Fig. 5 illustrates the distributional patterns of several types of dry beans according to their characteristics. Fig. 5(A) depicts the distributional pattern of area with different classes of dry bean Seker, Barbunya, Bombay, Cali, Horoz, Sira and Dermason by using boxplot.

Fig. 5(A) in feature area, the dry bean Dermason has a range of minimum and maximum values around fifteen thousand to forty-eight thousand respectively. The curve is symmetrical and this category has no outliers. Again, the dry bean Bombay class has minimum and maximum values roughly 130 thousand to 170 thousand correspondingly. Some of outliers are presented by dot-points exhibited in the top position. The distributional pattern of the data is mostly symmetrical. Fig. 5(A) shows the remaining boxplot which represents the dry bean Seker, Barbunya, Cali, Horoz and Sira with their minimum and maximum values and distributional patterns.

Fig. 5(B) illustrates the perimeter features within each type of dry beans. The bean Bombay represents a value of median almost 1580 and a maximum and minimum values approximately 1850 and 1300 pixels respectively. The distributional patterns of Bombay beans for the perimeter trait are mostly symmetrical with a few outliers visible outside the boxplot. The additional boxes in Fig. 5(B) illustrates the distributional patterns and outliers associated with the dry bean Seker, Barbunya, Cali, Horoz, Sira and Dermason. Similarly, additional Fig. 5(C) to Fig. 5(P) illustrate the statistical patterns with data distribution for the remaining features of the different dry beans including their outliers.

Fig. 6 shows the distributional patterns of the various categories of each variable with all descriptive statistics for the selected sixteen different features of dry bean represents in each boxplot. In particular, Fig. 5 displays the frequencies, IQR, and median of each category in this dry bean data set and represents outliers.

## 4. Experimental results

This section first describes details about confusion matrix and experimental setups. Then, the obtained results and performance of the proposed framework have been discussed and analyzed sequentially.

### 4.1. Confusion matrix for selected different algorithms

In this study, a confusion matrix is adopted to visualize and summarize the performance of the classifiers. The matrix clarifies the specific class accuracy of each dry bean as well as the incorrect classification rates. Each row of the confusion matrix represents the actual class while each column represents the predicted class. All the diagonal elements denote correctly classified outcomes and the off-diagonal elements of the matrix represent the misclassified outcomes. The confusion matrix is accomplished by XGB classifier using physical features such as form, shape, type and structure etc. The correct as well as confusing information of each class has been further clarified in Table 5. It reveals that with the exception of Dermason dry bean seeds, the frequency of all major diagonal elements is higher when the ADASYN algorithm is used than when it is not used in Table 5(a). Similarly, all other confusion matrices, Table 5(b) to Table 5(h) demonstrate the actual and predicted number of observations as listed in Table 5(a).

Table 6 shows the classification performance of the eight distinct models LR, KNN, DT, RF, SVM, NB, XGB, and MLP in the first column, and the performance measures AUC, ACC, MSE, $F_1$-score, FPR, Kappa, SE, and SP are presented in the top row, where upper one is without ADASYN and the another one is with ADASYN. When all classes ensure equal number of samples with applying ADASYN algorithm, the XGB classifier attains the highest performance measure, including an ACC that is more than 95% and an AUC that is 99.64%. Additionally, the rest of the performance measures are higher compared to other selected models when ADASYN is applied. The KNN (ACC 95%) and RF (ACC 94%) models show somewhat superior performance measures with re-

**Table 4**
Calculation formulas and explanations of multiple class metrics.

| Measure | Formula | Evaluation Focus |
|---|---|---|
| Averaged Accuracy | $\frac{\sum_{i=1}^{l} \frac{tp_i+tn_i}{tp_i+fp_i+tn_i+fn_i}}{l}$ | It is used to calculate the mean success of classes. |
| Averaged Error Rate | $\frac{\sum_{i=1}^{l} \frac{fp_i+fn_i}{tp_i+fp_i+tn_i+fn_i}}{l}$ | It is used to calculate the mean error rate of classes. |
| Averaged Recall ($rM$) | $\frac{\sum_{i=1}^{l} \frac{tp_i}{tp_i+fn_i}}{l}$ | It is used to calculate the mean of the remainder per class. |
| Averaged Precision ($pM$) | $\frac{\sum_{i=1}^{l} \frac{tp_i}{tp_i+fp_i}}{l}$ | It is used to calculate the mean of the precision per class. |
| $F1$-Score (Averaged F-Measure) | $\frac{2*pM*rM}{pM+rM} = \frac{2*\frac{\sum_{i=1}^{l} \frac{tp_i}{tp_i+fp_i}}{l}*\frac{\sum_{i=1}^{l} \frac{tp_i}{tp_i+fn_i}}{l}}{\frac{\sum_{i=1}^{l} \frac{tp_i}{tp_i+fp_i}}{l}+\frac{\sum_{i=1}^{l} \frac{tp_i}{tp_i+fn_i}}{l}}$ | It is used to calculate F1-score per class. |

**Table 5**
(a). Confusion matrix for XGBoost classifier with and without ADASYN algorithm.

| without ADASYN | | | | | | | |
|---|---|---|---|---|---|---|---|
| Actual | Predict Seker | Barbunya | Bombay | Cali | Horoz | Sira | Dermason |
| Seker | **454** | 6 | 0 | 0 | 0 | 14 | 13 |
| Barbunya | 0 | **316** | 0 | 22 | 2 | 2 | 1 |
| Bombay | 0 | 0 | **116** | 0 | 0 | 1 | 0 |
| Cali | 1 | 6 | 0 | **399** | 8 | 5 | 0 |
| Horoz | 0 | 3 | 0 | 11 | **466** | 8 | 7 |
| Sira | 8 | 2 | 0 | 2 | 4 | **558** | 62 |
| Dermason | 13 | 0 | 0 | 0 | 0 | 54 | **839** |

| with ADASYN | | | | | | | |
|---|---|---|---|---|---|---|---|
| Actual | Predict Seker | Barbunya | Bombay | Cali | Horoz | Sira | Dermason |
| Seker | **678** | 1 | 0 | 1 | 0 | 19 | 5 |
| Barbunya | 5 | **660** | 0 | 29 | 4 | 8 | 0 |
| Bombay | 0 | 0 | **710** | 0 | 0 | 0 | 0 |
| Cali | 5 | 8 | 0 | **702** | 8 | 3 | 0 |
| Horoz | 0 | 2 | 0 | 6 | **670** | 14 | 5 |
| Sira | 16 | 3 | 0 | 4 | 22 | **592** | 44 |
| Dermason | 17 | 0 | 0 | 0 | 4 | 63 | **625** |

**Table 6**
Performance measures of classification models (%) on the dry bean dataset.

| without ADASYN | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Classifiers | AUC | ACC | MS Error | F1-Score | FPR | Kappa | SE | SP |
| LR | 99.35 | 91.0 | 9.0 | 91.0 | 34.39 | 89.14 | 99.65 | 98.70 |
| KNN | 96.76 | 89.0 | 11.0 | 89.0 | 34.01 | 86.92 | 99.68 | 98.93 |
| DT | 94.85 | 90.0 | 10.0 | 90.0 | 34.64 | 87.33 | 99.04 | 98.46 |
| RF | 99.40 | 92.0 | 8.0 | 92.0 | 34.44 | 90.18 | 99.68 | 98.91 |
| SVM | 99. 58 | 92.0 | 8.0 | 92.0 | 34.40 | 90.67 | 99.68 | 98.50 |
| NB | 99.18 | 89.0 | 11.0 | 89.0 | 34.17 | 86.94 | 99.62 | 98.50 |
| **XGB** | **99.60** | **93.0** | **7.0** | **93.0** | **34.44** | **90.92** | **1.00** | **98.69** |
| MLP | 99.43 | 91.0 | 9.0 | 91.0 | 34.50 | 89.47 | 99.66 | 98.47 |

| with ADASYN | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Classifiers | AUC | ACC | MS Error | F1-Score | FPR | Kappa | SE | SP |
| LR | 98.03 | 83.0 | 17.0 | 83.0 | 34.25 | 80.60 | 96.96 | 97.45 |
| **KNN** | **98.52** | **95.0** | **5.0** | **95.0** | **33.62** | **92.77** | **99.85** | **99.50** |
| DT | 94.38 | 90.0 | 10.0 | 90.0 | 33.85 | 88.83 | 98.91 | 98.93 |
| **RF** | **99.60** | **94.0** | **6.0** | **94.0** | **33.68** | **92.62** | **99.10** | **99.70** |
| SVM | 98. 67 | 86.0 | 14.0 | 86.0 | 34.08 | 84.03 | 99.84 | 98.10 |
| NB | 97.16 | 79.0 | 21.0 | 79.0 | 34.47 | 75.87 | 95.31 | 97.05 |
| **XGB** | **99.64** | **95.4** | **6.9** | **94.0** | **33.66** | **93.10** | **99.92** | **99.85** |
| MLP | 98.38 | 84.0 | 16.0 | 84.0 | 34.19 | 81.74 | 98.70 | 96.95 |

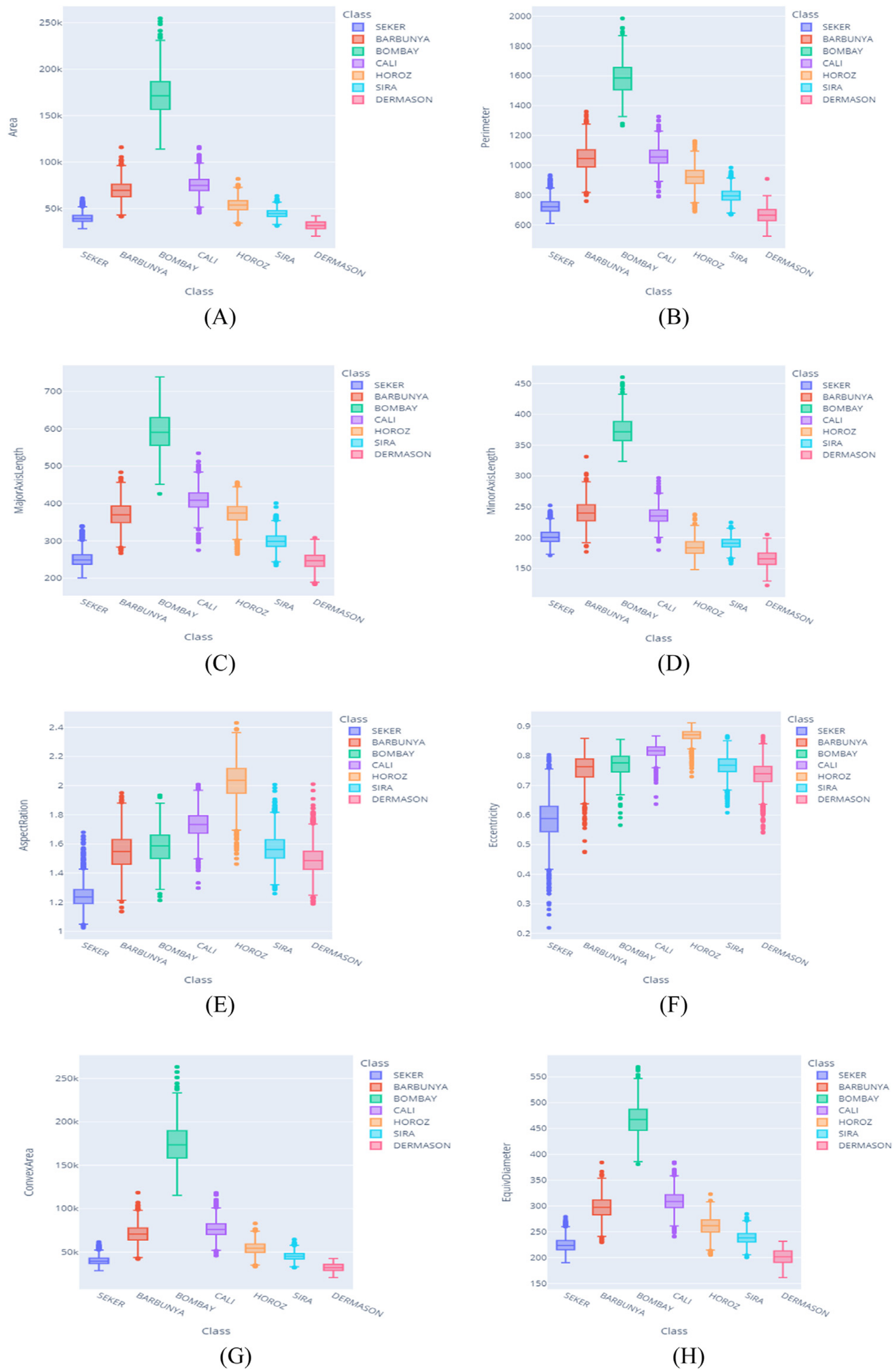


**Fig. 5.** (A-P) Distributional patterns of seven classes of dry bean among the features.

gard to accuracy metrics than LR (ACC 83%), SVM (ACC 86%), MLP (ACC 84%) with ADASYN algorithm in Table 6.

When compared to the performance of other models without ADASYN algorithm, the XGB model provides more accuracy metrics. The ACC and AUC of the XGB model are shown in Table 6 and it is 93% and more than 99% respectively. The performance accuracy of LR (ACC 91%), DT (ACC 90%), RF (ACC 92%) and MLP (ACC 91%) models are higher than that of KNN (ACC 89%) and NB (ACC 89%). We conclude from these discussions that the XGB model demonstrates better performance when compared to other models with or without ADASYN algorithm. In addition, the KNN model (ACC 95%) achieves the second highest accuracy in terms of utilizing the ADASYN algorithm, while the RF model (ACC 92%) has the highest average accuracy in terms of not using the ADASYN algorithm.

When compared with the same model, XGB model has better performance in both cases of with or without ADASYN algorithms. Similarly, the XGB model has better performance in terms of ACC and AUC, which are 95% and 93% respectively with and without ADASYN ap-

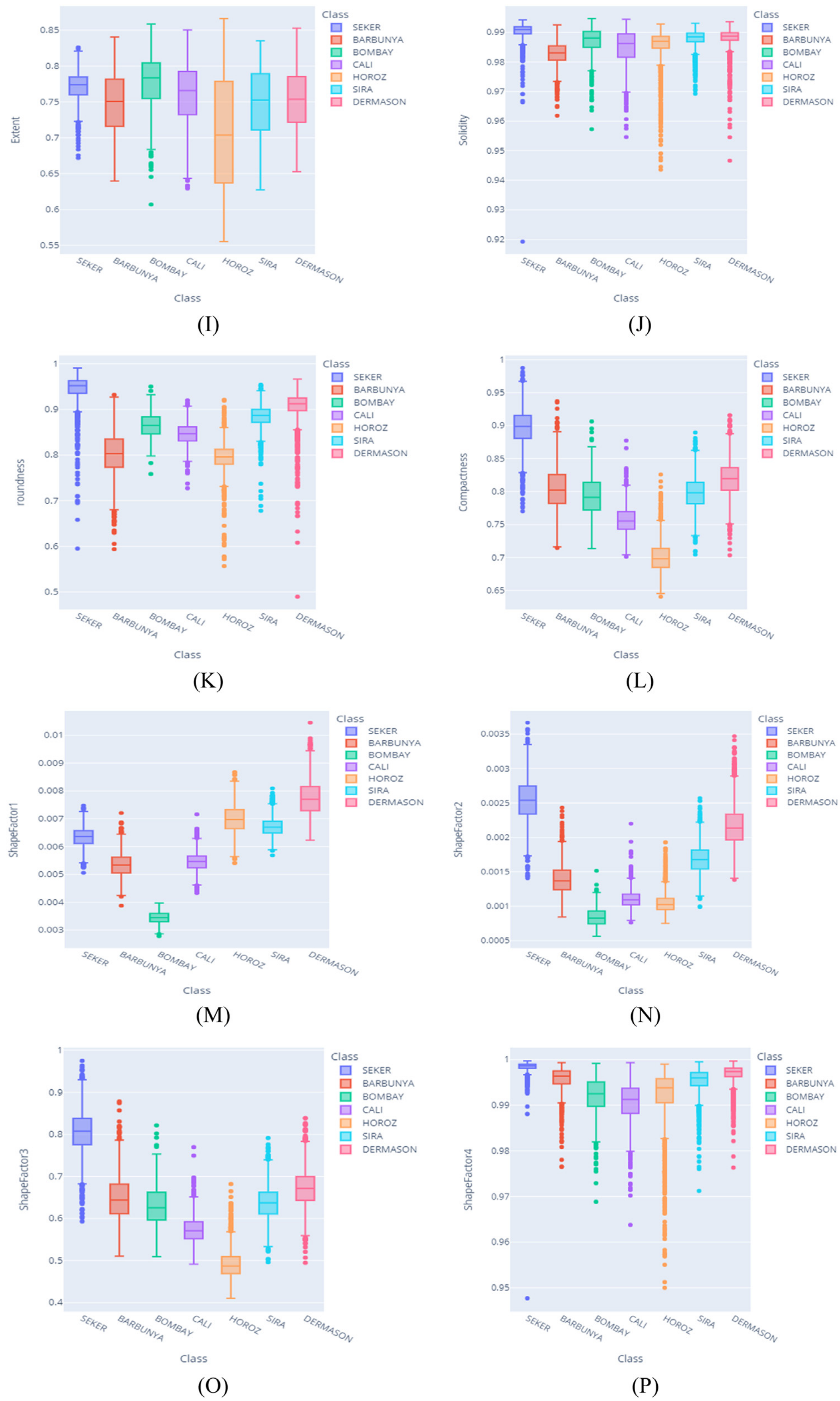


**Fig. 5.** Continued

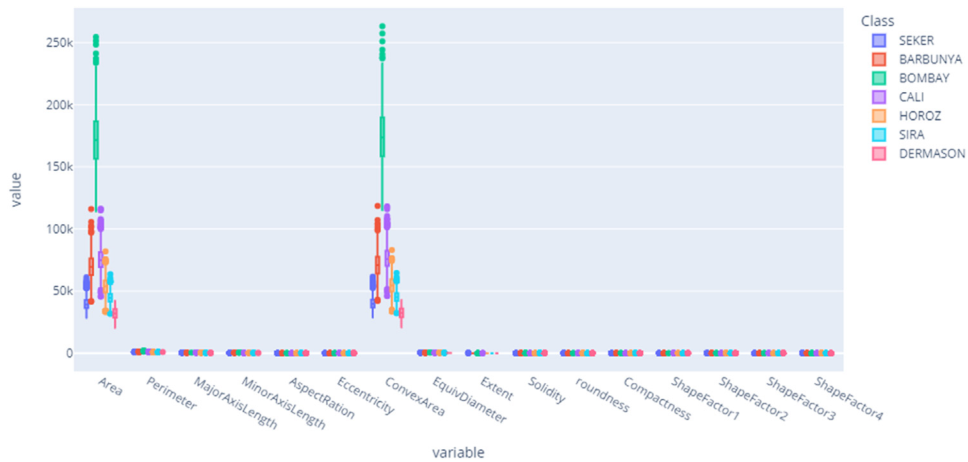


**Fig. 6.** Box-plot for different features with its different classes.

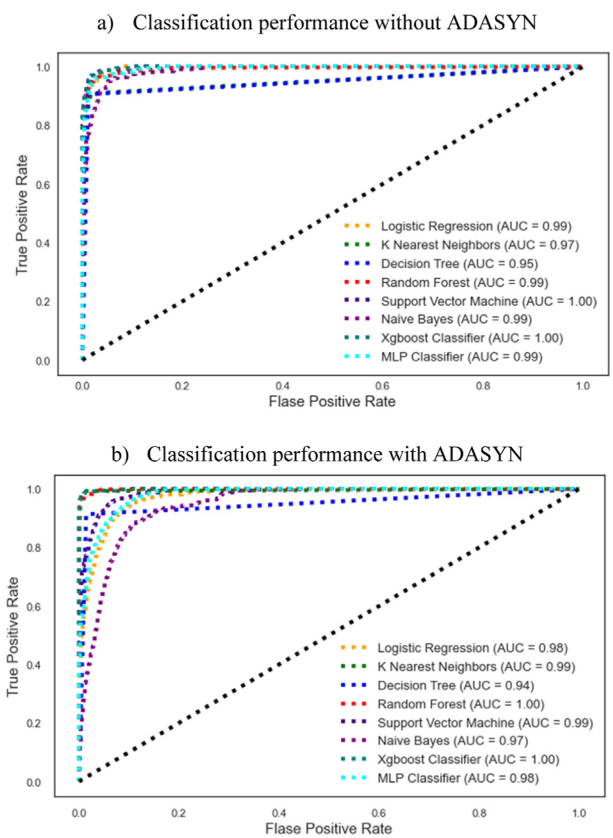


**Fig. 7.** ROC curve for checking performance of the selected models.

proaches. The KNN model shows good performance at the time of applying ADASYN algorithm, but it presents a relatively poor performance in the absence of ADASYN method. For RF classification model, the results remain almost identical with accuracy 91% in both cases of ADASYN algorithm. Most of the models show better performance with the usage of ADASYN algorithm. In contrast, MLP with ADASYN algorithm obtains 84% accuracy but not as good as the absence of ADASYN algorithm with 91% accuracy. Similarly, the performance of NB is the lowest accuracy of 79% in the presence of ADASYN and of 89% without ADASYN. For this dry bean dataset, the XGB model provides the excellent performance in both cases with and without ADASYN algorithms.

Fig. 7 presents the models' performance comparison among the chosen classifiers with and without using ADASYN algorithm. Fig. 7(a) illustrates the performance of classifiers without utilizing ADASYN where it is impossible to prove that which classifier is superior performer for this dry bean dataset. In this figure, the AUC value for both XGB and SVM models are 1. Furthermore, the color of TPR and FPR for the classifier XGB classifier has the highest point in Fig. 7(a), whereas the colors of the lines for LR, RF, and MLP models are comparatively better with AUC values of 0.99.

Fig. 7(b) depicts the performance of the classifiers applying ADASYN algorithm, where XGB and RF exhibit the greatest performance in terms of their AUC values as well as the classification thresholds that means TPR vs. FPR in ROC curve. Similarly, the difference between TPR and FPR of the ROC curve for the NB model indicates its somewhat worse classification performance. The KNN and SVM models of Fig. 7(b) have also indicated relatively superior accuracy measures, whereas the classification thresholds have continuously increased with the value of AUC. For this reason, it is very difficult to determine which classifier is superior among KNN, RF, SVM and XGB when applied with ADASYN. The boxplots are used in Fig. 9(a & b) to compare the classification performance of the chosen models.

Fig. 8 represents the accuracy rate of eight distinct models for all types of beans. The Bombay type of bean is classified more precisely by the selected models, except for multilayer perception, while Sira is classified with the least accuracy. The XGB approach performs better for all dry bean classes with accuracies of 95% in Seker, 97% in Barbunya, 100% in Bombay, 97% in Cali, 96% in Horoz, 85% in Sira and 99% in Dermason compared to other models. In some of classes, KNN performs almost same with XGB in Bombay and Dermason.

After optimizing parameters, correct performance metrics are shown in Fig. 9 with two scenarios: a) without ADASYN algorithm and b) with ADASYN algorithm. As shown in Figs. 9(a) and 9(b), the XGB classifier shows superior performance compared to other models by achieving more than 0.96 accuracy value. All of the discussions about classification performance demonstrate that the XGB classifier outperforms among all other selected models in both with and without utilizing ADASYN algorithm.

Categorizations of dry beans are important for agricultural as well as business sectors in the winter-based nations. In this research, there have been seven distinct varieties of dry beans, and eight types of classifiers are applied for identification of dry bean. To evaluate the feature properties for bean identification, we have used a vertical bar-diagram for each feature representation in Fig. 10 using XGB technique**.** The XGB model shows better performance in the cases of ADASYN and without ADASYN algorithms among the selected classifiers**,** that is why the authors apply it to represent the feature importance**.** ShapeFactor2, Minor Axis Length

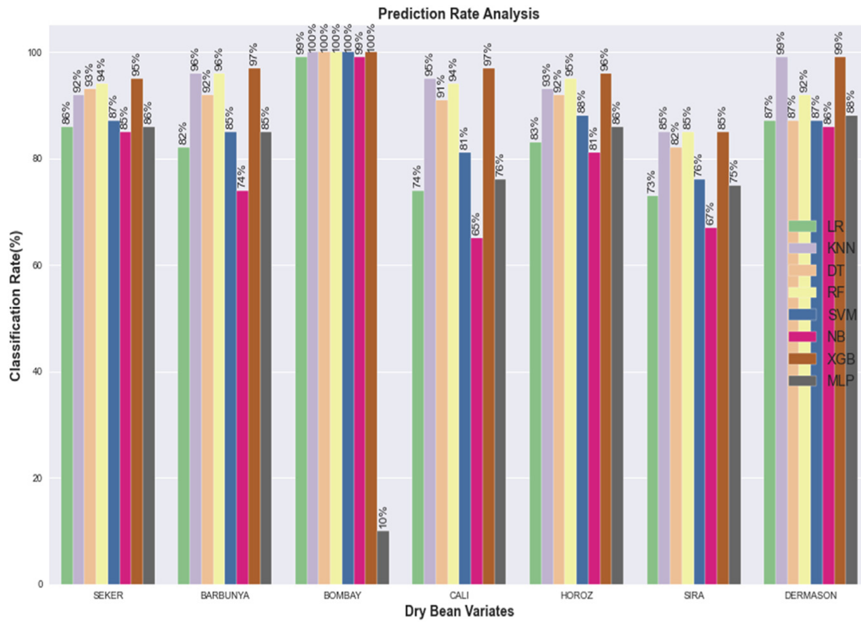


**Fig. 8.** Performance of classifiers for each type of dry bean.

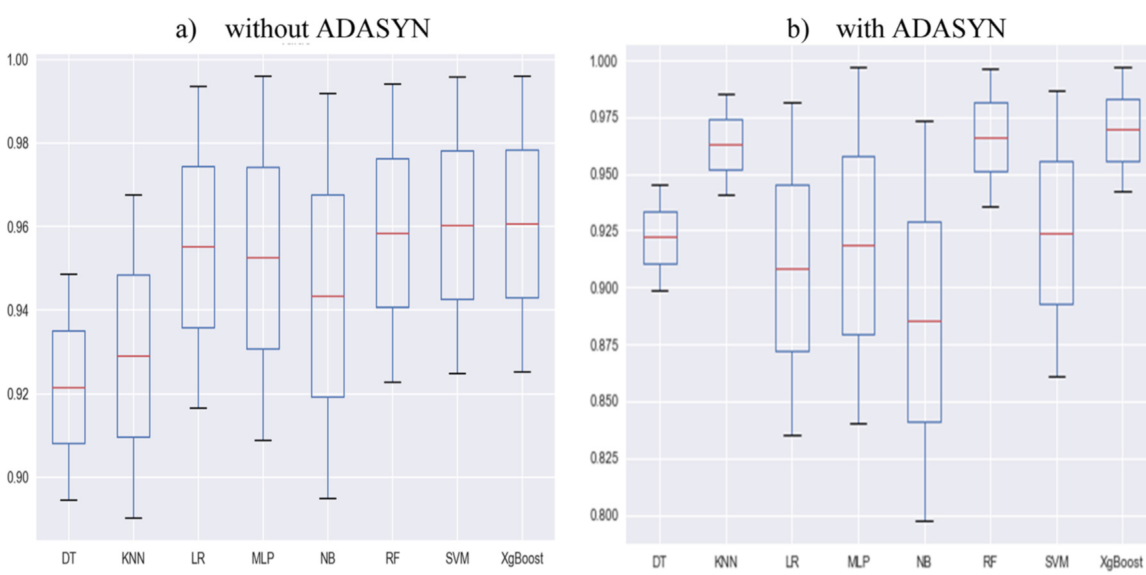


**Fig. 9.** Comparison of selected models with and without ADASYN algorithm.

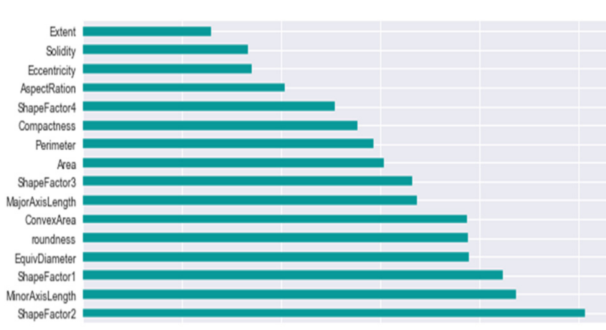


**Fig. 10.** Feature importance by using XGBoost algorithm.

and ShapeFactor1 are the most essential criteria in the classification of this dry bean, while EquivDiameter, Roundness and ConvexArea are nearly equally relevant. However, the extent, solidity, eccentricity, and AspectRatio have minimum value during dry bean classification.

## 5. Discussion

In this framework, several machine learning techniques are applied so that the dry beans from crop production can be classified uniformly with low computational cost as well as overcome the bean intra-class variations. Recently, a large number of researchers have established different algorithms mostly for balanced datasets. The skewed data points pose difficulties in multi-class problem classification, particularly multiclass imbalanced data. These existing drawbacks of classification may be handled by making equal distribution of features in all classes with the help of ADASYN algorithm.

The suggested technique has been evaluated with the following metrics such as ACC, SE, SP, Kappa, FPR and MSE. The XGB model surpasses

**Table 7**
Performance comparison among proposed models and other' studies.

| Name of authors | Dataset | Selected models | Best model | Accuracy (%) |
|---|---|---|---|---|
| Koklu et al. (2020) | Dry bean seeds | MLP, SVM, DT, and KNN | SVM | 93.13 |
| Sanlı et al. (2020) | Sugar beet seeds | MSI, MIT | MIT | 82.0 |
| Kiratiratanapruk et al. (2020) | 14 kinds of rice seeds | LR, LDA, KNN, CNN and SVM | SVM | 83.9 |
| Pozza *et al.*(2022) | Bean seeds | RF, rpart, rpart1SE, rpart2 NB, SVM | RF | 80.0 |
| Keya et al. (2020) | 5 variants of seed | CNN | CNN | 93.0 |
| Słowiński, G. (2021) | Dry bean seeds | NB, SVM, DT, RF, SVC, and ANN | RF | 93.61 |
| OuYang et al. (2010) | Rice seeds | BP-ANN | BP-ANN | 93.66 |
| **Proposed model** | **Dry bean seeds** | LR, KNN, DT, RF SVM, NB, XGB and MLP. | **XGB without ADASYN** | **93.00** |
| | | | **XGB with ADASYN** | **95.40** |

all other tree-based classifiers on the imbalanced dry beans dataset in terms of ACC, SE, SP and AUC. Additionally, these models provide the higher accuracy for this dry bean dataset than other metrics. According to the experimental assessments, XGB classifier outperforms both with and without ADASYN. Fig. 7(b) shows that KNN and RF classifiers perform equally well but not as good as XGB classifier. The performance of KNN is slightly better than RF. The NB, LR and SVM perform at the cost of a much higher run time. The investigation shows that the multiclass classification model may enhance performance by around 1.87%, when oversampled with ADASYN method.

The performances of our multiclass classification model as well as prior competing methods on various datasets are enlisted in Table 7. It is not possible to directly compare our method with those of the existing models in terms of precision, recall and accuracy. Since the prior approaches have followed traditional strategies in seed preprocessing stage, feature extraction and classification with different distributions of experimental dataset. For example, the accuracy is 83.90% for 14 kinds of rice seeds including statistical machine learning approaches and pre-trained models on deep learning techniques (Kiratiratanapruk et, al.,2020) and 93.00% on 5 variants of seed, developing an adequate integrated framework to replace the current classification system (Keya et, al.,2020).

The previous approaches fail to reflect the ability of the classifier for each class of samples because of unbalanced data set as well as favoring each class with higher probability of occurrence (majority) over another with low probability (minority) of occurrence (Awal *et al.*, 2021a,2021b). Oliveira et al. experimented on a total of 1800 beans with four classes and observed that the imbalanced dataset represents true variation of class. The classification accuracies were 0.93% for unbalanced dataset and 0.92% for balanced dataset but precision, recall and $F_1$-score were also high for few classes. As those classes are classified inaccurately and affect the performance metrics for their respective class due to improper distribution. The RF classification model in balanced dataset provided more information, as it ignores the effects of the number of samples from an input class to influence the accuracy of classification (Oliveira *et al.*, 2021).

As mentioned earlier, we have used 13,611 items with 7 different types of dry beans collected from various planting areas in Turkey under varying imaging conditions in our experimental evaluations. Additionally, adaptive synthetic sampling technique is adopted to eliminate the issues occurred due to imbalanced dataset and made an attempt to delineate how the algorithm affects the classification performance when working at high production volumes. Only the proposed approaches (Słowiński *et al.* 2021; Pozza et al., 2022) experimented on dry beans with a large number of images and subjects similar to this work. The comparison with state-of-the-art algorithms demonstrates that the XGB classification model shows good performance with possible highest accuracy when all classes have equal number of samples. However, our established model has the lowest possibility to be affected by different levels of noise implications in geometric feature fusion cases.

### 5.1. Reasons of better performance behind the XGB classifier

The machine learning algorithm with XGB classifier performs better due to gradient boosting, minimizing loss function, and avoiding overfitting. In the additive and sequential stages, the trees are generated in sequential approach which turns the weak learners into strong learners by adding up weights to the weak learners as well as decrease the weights of strong learners. In the similar process, every tree boosts and learns consistently from the prior tree grown. Another advantage is to overcome the tiring process using approximate greedy algorithm by dividing the invariant data into quantiles or adopts quantiles as candidate thresholds to split. The parameters are the key factors behind the better performance of the classifier.

### 5.2. Strengths, limitations, and future scopes of the study

The main strength of this study is to identify automatically uniform seed varieties for more crop production with reducing computational cost as well as overcoming the inter-seed ambiguities. Due to increasing demand of uniform seeds in agricultural fields, the proposed technique may be applied to determine seed quality for planting and marketing, and classification. The investigated method shows better performance to identify the accurate types of dry beans.

The experiments are conducted on secondary data, which consists of 7 different types of dry beans. Nearly the 13,611 items of the dataset were collected from various planting areas of bean by the research institute in Turkey. The performance of the developed algorithm may be reduced with respect to other datasets with poor data pre-processing and segmentation, types of features and feature dimensionality and so on.

Although, the classifiers have achieved a satisfactory accuracy for this dataset, it still suffers from various real-time challenges in uniform bean identification. Here, only the variables related to shape and size, and characteristics of bean cultivar are included as features. The suture axis of bean (i.e., third dimensional analysis) is ignored due to huge time consuming, but can increase the classification performance. If the coefficient of variance e.g., the difference in the shape of each bean variety is considered in shape and size variables, it will improve accuracy. The bean identification performance may be enhanced by employing feature fusion namely shape and size features, texture features, statistical features and decision level fusion.

## 6. Conclusions

This work studies the classification performance with imbalanced and balanced distribution of dry bean dataset which has a great impact on data science and agricultural fields. The agricultural products highly depend on the quality of seeds as well as the fertility of lands. In this study, a genetically diversed dry bean dataset is used to identify the actual seeds with the help of ADASYN algorithm and different machine learning techniques which are described details in classification stage. The mentioned techniques have showed different performances with various parameter settings. Among them, the XGB beats all other approaches with both balanced and imbalanced classes for the experimental dataset with ACC of 93% and 95% respectively. In the case of a balanced dataset, KNN and RF algorithms also demonstrate superior performance in terms of accuracy such as ACC, SE, SP and Kappa among the others. ShapeFactor2, MinorAxis Length, ShapeFactor1, EquivDiameter, roundness and Convex Area are most important features to determine these dry beans. The existing classification approaches employ multi-class classification algorithm which is highly recommended for isolation and improvement of unified dry bean successfully. Before applying machine learning algorithms to identify important predictive factors and the classes of bean, this study suggests that the authors should investigate the distribution of datasets in distinct classes with focusing on their balanced or imbalanced patterns.

## Funding

The author(s) received no financial support for the research, authorship, and/or publication of this paper.

## Declaration of Competing Interest

There are no conflicts of interest.

## Acknowledgments

The authors gratefully acknowledge the contribution of Statistics Discipline, Science, Engineering and Technology School, Khulna University, Khulna-9208, Bangladesh. We give special thanks to Milon Sheikh, a student of English Discipline, Khulna University. He has tried his best to improve the language issues of this article. The authors also thank and gratefully acknowledge to the editor and referees for their comments and positive critique.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijcce.2023.01.002.

## References

Agresti, A. (2002). *Categorical data analysis* (2nd ed). New York: Wiley.

Ahmed, N., Ahammed, R., Islam, M. M., et al., (2021). Machine learning based diabetes prediction and development of smart web application. *International Journal of Cognitive Computing in Engineering, 2*, 229–241.

Alzubi, J. A., Kumar, A., Alzubi, O., & Manikandan, R. (2019). Efficient approaches for prediction of brain tumor using machine learning techniques. *Indian Journal of Public Health Research & Development, 10*(2).

Alzubi, O. A., Alzubi, J. A., Alweshah, M., et al., (2020). An optimal pruning algorithm of classifier ensembles: Dynamic programming approach. *Neural Computing and Applications, 32*(20), 16091–16107. 10.1007/s00521-020-04761-6.

Alzubi, O. A., Alzubi, J. A., Al-Zoubi, A. M., et al., (2021). An efficient malware detection approach with feature weighting based on Harris Hawks optimization. *Cluster Computing*, 1–19. 10.1007/s10586-021-03459-1.

Andhalkar, S., & Momin, B. F. (2018, July). Multiclass IFROWNN classification algorithm using OVA and OVO strategy. In *2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT)* (pp. 1–7). IEEE.

Awad, M., & Khanna, R. (2015). Support vector regression. In *Efficient learning machines* (pp. 67–80). Berkeley, CA: Apress.

Awal, M. A., Hossain, M. S., Debjit, K., et al., (2021a). An early detection of asthma using BOMLA detector. *IEEE Access : Practical Innovations, Open Solutions, 9*, 58403–58420.

Awal, M. A., Masud, M., Hossain, M. S., et al., (2021b). A novel bayesian optimization-based machine learning framework for COVID-19 detection from inpatient facility data. *IEEE Access : Practical Innovations, Open Solutions, 9*, 10263–10281.

Bacchetta, G., Grillo, O., Mattana, E., & Venora, G. (2008). Morpho-colorimetric characterization by image analysis to identify diaspores of wild plant species. *Flora-Morphology, Distribution, Functional Ecology of Plants, 203*(8), 669–682.

Barbon, A. P. A., Barbon Jr,, S., Mantovani, R. G., et al., (2016). Storage time prediction of pork by Computational Intelligence. *Computers and Electronics in Agriculture, 127*, 368–375.

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32.

Burges, C. J. (1998). A tutorial on support vector machines for pattern recognition. *Data mining and Knowledge Discovery, 2*(2), 121–167. 10.1023/A:1009715923555.

Ceyhan, E., Kahraman, A., & Onder, M. (2012). The impacts of environment on plant products. *International Journal of Bioscience, Biochemistry and Bioinformatics, 2*(1), 48.

Desai, C., Ramanan, D., & Fowlkes, C. C. (2011). Discriminative models for multi-class object layout. *International Journal of Computer Vision, 95*(1), 1–12.

Gómez-Sanchis, J., Martín-Guerrero, J. D., Soria-Olivas, E., et al., (2012). Detecting rottenness caused by Penicillium genus fungi in citrus fruits using machine learning techniques. *Expert Systems with Applications, 39*(1), 780–785.

Goseva-Popstojanova, K., Anastasovski, G., & Pantev, R. (2012, November). Using multiclass machine learning methods to classify malicious behaviors aimed at web systems. In *2012 IEEE 23rd International Symposium on Software Reliability Engineering* (pp. 81–90). IEEE.

Granitto, P. M., Garralda, P. A., Verdes, P. F., & Ceccatto, H. A. (2002). Boosting classifiers for weed seeds identification. *VIII congreso argentino de ciencias de la computación*.

Gürcan, F. (2018). Multi-class classification of turkish texts with machine learning algorithms. In *2018 2nd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)* (pp. 1–5). IEEE.

Hastie, T., Tibshirani, R., Friedman, J. H., et al., (2009). *The elements of statistical learning: Data mining, inference, and prediction* (pp. 1–758). New York: Springer. Vol. 2.

Igual, L., & Seguí, S. (2017). Introduction to data science. In *Introduction to data science* (pp. 1–4). Cham: Springer.

Islam, M. M., Rahman, M. J., Islam, M. M., et al., (2022). Application of machine learning based algorithm for prediction of malnutrition among women in Bangladesh. *International Journal of Cognitive Computing in Engineering, 3*, 46–57.

Islam, M., Dinh, A., Wahid, K., & Bhowmik, P. (2017, April). Detection of potato diseases using image segmentation and multiclass support vector machine. In *2017 IEEE 30th canadian conference on electrical and computer engineering (CCECE)* (pp. 1–4). IEEE.

Keya, M., Majumdar, B., & Islam, M. S. (2020). A Robust Deep Learning Segmentation and Identification Approach of Different Bangladeshi Plant Seeds Using CNN. In *2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT)* (pp. 1–6).

Khan, W., & Haroon, M. (2022). An unsupervised deep learning ensemble model for anomaly detection in static attributed social networks. *International Journal of Cognitive Computing in Engineering, 3*, 153–160.

Khatri, A., Agrawal, S., & Chatterjee, J. M. (2022). Wheat Seed Classification: Utilizing Ensemble Machine Learning Approach. *Scientific Programming*, 2022.

Kiratiratanapruk, K., Temniranrat, P., Sinthupinyo, W., et al., (2020). Development of paddy rice seed classification process using machine learning techniques for automatic grading machine. *Journal of Sensors*, 2020.

Koklu, M., & Ozkan, I. A. (2020). Multiclass classification of dry beans using computer vision and machine learning techniques. *Computers and Electronics in Agriculture, 174*, Article 105507. 10.1016/j.compag.2020.105507.

Kumari, S., Kumar, D., & Mittal, M. (2021). An ensemble approach for classification and prediction of diabetes mellitus using soft voting classifier. *International Journal of Cognitive Computing in Engineering, 2*, 40–46.

Lang, T., Flachsenberg, F., von Luxburg, U., et al., (2016). Feasibility of active machine learning for multiclass compound classification. *Journal of Chemical Information and Modeling, 56*(1), 12–20.

Lawi, A., & Adhitya, Y. (2018). Classifying physical morphology of cocoa beans digital images using multiclass ensemble least-squares support vector machine. *Journal of Physics: Conference Series*. IOP Publishing Vol. 979, No. 1.

León-Roque, N., Abderrahim, M., Nuñez-Alejos, L., et al., (2016). Prediction of fermentation index of cocoa beans (Theobroma cacao L.) based on color measurement and artificial neural networks. *Talanta, 161*, 31–39.

Li, B. (2021). Hearing loss classification via AlexNet and extreme learning machine. *International Journal of Cognitive Computing in Engineering, 2*, 144–153.

Liu, J., Yang, W. W., Wang, Y., et al., (2011). Optimizing machine vision-based applications in agricultural products by artificial neural network. *International Journal of Food Engineering, 7*(3).

Madhu, B., Mukherjee, A., Islam, M. Z., et al., (2021, December). Depth motion map based human action recognition using adaptive threshold technique. In *2021 5th International Conference on Electrical Information and Communication Technology (EICT)* (pp. 1–6). IEEE.

Mendoza, F., Dejmek, P., & Aguilera, J. M. (2010). Gloss measurements of raw agricultural products using image analysis. *Food Research International, 43*(1), 18–25.

Movassagh, A. A., Alzubi, J. A., Gheisari, M., et al., (2021). Artificial neural networks training algorithm integrating invasive weed optimization with differential evolutionary model. *Journal of Ambient Intelligence and Humanized Computing*, 1–9. 10.1007/s12652-020-02623-6.

Mukherjee, A., Islam, M. Z., Mamun-Al-Imran, et al., (2021, September). Iris recognition using wavelet features and various distance based classification. In *2021 International Conference on Electronics, Communications and Information Technology (ICECIT)* (pp. 1–4). IEEE.

Mukherjee, A., Ripon, K. S. N., Ali, L. E., et al., (2022). Image gradient based iris recognition for distantly acquired face images using distance classifiers. In *International Conference on Computational Science and Its Applications* (pp. 239–252). Cham: Springer.

Müller, K. R., Mika, S., Tsuda, K., et al., (2018). An introduction to kernel-based learning algorithms. *Handbook of neural network signal processing*. CRC Press 4-1.

Muralidharan, K., Ramesh, A., Rithvik, G., et al., (2021). 1D Convolution approach to human activity recognition using sensor data and comparison with machine learning algorithms. *International Journal of Cognitive Computing in Engineering, 2*, 130–143.

Oliveira, M. M., Cerqueira, B. V., Barbon Jr,, S., et al., (2021). Classification of fermented cocoa beans (cut test) using computer vision. *Journal of Food Composition and Analysis, 97*, Article 103771.

Paliwal, J., Visen, N. S., & Jayas, D. S. (2001). Evaluation of neural network architectures for cereal grain classification using morphological features. *Journal of Agricultural Engineering Research, 79*(4), 361–370.

Pozza, E. A., de Carvalho Alves, M., & Sanches, L. (2022). Using computer vision to identify seed-borne fungi and other targets associated with common bean seeds based on red–green–blue spectral data. *Tropical Plant Pathology, 47*(1), 168–185.

Przybył, K., Gawałek, J., Koszela, K., et al., (2018). Artificial neural networks and electron microscopy to evaluate the quality of fruit and vegetable spray-dried powders. Case study: Strawberry powder. *Computers and Electronics in Agriculture, 155*, 314–323.

Rai, N., Kaushik, N., Kumar, D., et al., (2022a). Mortality prediction of COVID-19 patients using soft voting classifier. *International Journal of Cognitive Computing in Engineering, 3*, 172–179.

Rai, N., Kumar, D., Kaushik, N., et al., (2022b). Fake News Classification using transformer based enhanced LSTM and BERT. *International Journal of Cognitive Computing in Engineering, 3*, 98–105.

Rathi, M., & Pareek, V. (2016). Hybrid approach to predict breast cancer using machine learning techniques. *International Journal of Computer Science Engineering, 5*(3), 125–136.

Rehman, T. U., Mahmud, M. S., Chang, Y. K., et al., (2019). Current and future applications of statistical machine learning algorithms for agricultural machine vision systems. *Computers and Electronics in Agriculture, 156*, 585–605.

Rodríguez-Pulido, F. J., Gordillo, B., González-Miret, M. L., & Heredia, F. J. (2013). Analysis of food appearance properties by computer vision applying ellipsoids to colour data. *Computers and Electronics in Agriculture, 99*, 108–115.

Sáez, A., Sánchez-Monedero, J., Gutiérrez, P. A., & Hervás-Martínez, C. (2015). Machine learning methods for binary and multiclass classification of melanoma thickness from dermoscopic images. *IEEE transactions on medical imaging, 35*(4), 1036–1045.

Sanlı, T., Sıcakyüz, Ç., & Yüregir, O. H. (2020). Comparison of the accuracy of classification algorithms on three data-sets in data mining: Example of 20 classes. *International Journal of Engineering, Science and Technology, 12*(3), 81–89.

Savakar, D. (2012). Identification and classification of bulk fruits images using artificial neural networks. *International Journal of Engineering and Innovative Technology (IJEIT), 1*(3), 35–40.

Sethuraman, J., Alzubi, J. A., Manikandan, R., et al., (2019). Eccentric methodology with optimization to unearth hidden facts of search engine result pages. *Recent Patents on Computer Science, 12*(2), 110–119. 10.2174/2213275911666181115093050.

Słowiński, G. (2020). Dry Beans Classification Using Machine Learning Multinomial Naive Bayes classifier. In *29th International Workshop on Concurrency, Specification and Programming (CS&P21), University of Technology and Economics, ul. Jagiellońska 82f, 03-301* (p. 2020).

Stegmayer, G., Milone, D. H., Garran, S., & Burdyn, L. (2013). Automatic recognition of quarantine citrus diseases. *Expert Systems with Applications, 40*(9), 3512–3517.

Subasi, A. (2015). A decision support system for diagnosis of neuromuscular disorders using DWT and evolutionary support vector machines. *Signal, Image and Video Processing, 9*(2), 399–408.

Yahyaoui, A., & Yumuşak, N. (2018). Decision support system based on the support vector machines and the adaptive support vector machines algorithm for solving chest disease diagnosis problems.

Zhang, C., Liu, C., Zhang, X., & Almpanidis, G. (2017). An up-to-date comparison of state-of-the-art classification algorithms. *Expert Systems with Applications, 82*, 128–150.

Zhang, L., & Zhan, C. (2017, May). Machine learning in rock facies classification: An application of XGBoost. In *International Geophysical Conference, Qingdao, China, 17-20 April 2017* (pp. 1371–1374). Society of Exploration Geophysicists and Chinese Petroleum Society.